IMPERIAL SCHOOL OF BANKING AND MANAGEMENT STUDIES · PUNE

Capstone Project Expectations

Agentic AI & Advanced Analytics in Finance

Post Graduate Diploma in Management (PGDM) in Financial Management

Semester III · Batch 2025–2027 · Course Code PGDM-SEM3-SPEC-AIFINANCE

87 project groups · 174 students · 2 credits · 30 hours

Faculty: Tejas Jadhav, CFA, FRM

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Deliverables

You submit two things and sit one viva. Everything else in this document explains how to build them.

Deliverable	What it is	When	Marks
1. Report	25 to 30 typed pages in the structure set out in section 4. Times New Roman 12, headings 14, line spacing 1.5, justified, one-inch margins, spiral bound. Similarity index 20 per cent or lower.	Hard copy and PDF, before the End Semester Examination	20
2. PowerPoint deck	Eight to ten slides covering your objectives, data sources, analysis and findings. **The viva is conducted on this deck.**	Bring it to the viva	Part of the viva
3. Viva voce	You present your PowerPoint deck to a panel of one internal faculty member and one external examiner. It includes a three-minute live demonstration of your system that shows one failure.	On your viva date	30

> **Total: 50 marks.** The report proves the work. The deck and the live demonstration prove that you did it yourself.

1. What the Capstone Is

The capstone project applies what you learned in this course to a real financial question, using data that is already published. You work in the pair you were allotted. Your project title is fixed and appears against your names in section 6 of this document.

Three rules govern everything that follows, and all three come from the institute's *SIP and Capstone Project Report Guidelines 2026*.

Rule	What it means for you
Secondary data only	Every number in your report comes from a published source. No surveys, no interviews, no company-supplied ledgers. If your project needs data nobody publishes, you cannot do it, and the topics in section 6 have already been checked against this.
25 to 30 pages	A typed report in the structure set out in section 4. Times New Roman 12, headings 14, line spacing 1.5, justified, one inch margins, spiral bound.
Submitted before the End Semester Examination	Hard copy and PDF. No extension is granted beyond this date, and late submissions are not accepted.

> **Academic integrity.** Similarity index must be 20 per cent or lower. Every data source and reference is acknowledged. Data fabrication or plagiarism means disqualification from evaluation. If a number in your report cannot be traced to a source in your bibliography, remove it.

Why the titles were adjusted

Seventy of the eighty-seven titles are exactly as the students submitted them. Thirteen were replaced because they failed one of the rules above: four needed data that nobody publishes, five had no financial calculation to defend at the viva, three were calculators that produce no finding, and one was not a finance topic. Three groups submitted nothing and were allotted a project. One group submitted two titles and now has one. In twelve of the replaced cases the subject area was kept, and group 70 later proposed a new finance topic of its own. What changed is the question the project answers.

2. How You Are Marked

The capstone carries fifty marks in total. Twenty come from the report and thirty from the viva voce, conducted by a panel of one internal faculty member and one external examiner.

Report — 20 marks

Criterion	Sub-criteria	Marks
Foundations and Reliability	Data and ethics, methodological and sampling choice, formatting and ethics	5
Technical Execution and Logic	Conceptual clarity, theoretical fit, analysis accuracy and interpretation	**10**
Strategic Value and Recommendation	Practicality, financial feasibility, future impact	5

Viva Voce — 30 marks

Criterion	Sub-criteria	Marks
Verification of Authenticity and Ownership	Data provenance, methodological choice, tool proficiency	5
Technical Depth and Financial Logic	Understanding the variables, assumption defence, formula mechanics	**10**
Integration with Macro-Environment	Current affairs linkage, regulatory knowledge, sectoral awareness	5
Behavioural and Professional Conduct	Confidence under pressure, precision of speech, critical self-awareness	10

> **Read the two bold rows together.** Twenty of the fifty marks test whether you can explain your own calculation and defend the assumptions inside it. Software that runs but that you cannot explain line by line earns neither. Build the analysis yourself and know why every number is what it is.

You also prepare a PowerPoint presentation of eight to ten slides covering objectives, data sources, analysis and findings. The viva is conducted on this deck.

3. The System You Build

This course produced nine working outputs across ten modules. Your capstone is not a new project built from nothing. It is those outputs assembled into one system that answers your question. Build it in five layers, because each layer earns a specific line of the marking scheme.

Layer	What it does	Files	Marks it earns
L1 Ingest	Pulls every input from a named public source and logs where each one came from	data/, ingest.py, sources.csv	Data and ethics (5) · Data provenance
L2 Extract	Pulls structure out of filings, or indexes documents for retrieval, or both	extract.py, rag/	Methodological choice
L3 Engine	The financial mathematics. Plain Python, no model in the loop, tested against a hand-worked example	engine/, tests/	**Technical execution (10) · Technical depth (10)**
L4 Commentary	Writes over the engine's output only, and cites every claim back to L1 or L3	commentary.py, governance.md	Tool proficiency · Regulatory linkage (5)
L5 Output	An application the panel can open, plus a generated document that becomes an annexure	app.py, report.py	Strategic value (5)

Ten rules

#	Rule	Why
1	Build all five layers. A project with no L3 engine is a demonstration, not a capstone.	L3 carries twenty of the fifty marks.
2	The mathematics lives in Python, never in the language model.	A language model works on language, not arithmetic. A number it produced cannot be defended under formula mechanics.
3	Every generated sentence cites a source or a computed number.	Uncited generated text is the easiest thing for an external examiner to break.
4	sources.csv records the URL, the publisher and the date pulled, for every input.	The viva's first criterion is verification of authenticity. This file is your evidence.
5	Every engine ships with a test against a hand-worked example.	It turns "my model says" into "my model reproduces the published figure".
6	State your assumptions in one file and defend them in the viva.	Assumption defence is named in the rubric.
7	Where a ledger does not exist publicly, use synthetic data and say so on the methodology page.	Declaring it costs nothing. Concealing it fails data provenance.
8	Any tool that outputs buy, sell or hold carries an educational-use disclaimer.	SEBI's Research Analyst and Investment Adviser regulations govern advice issued to the public.
9	Keep the work in a public repository with commits across the project period.	A single commit the night before submission is an ownership problem.
10	Your viva demonstration runs in three minutes and includes one failure.	Showing the system reject a bad input proves it is real.

4. The Report, Chapter by Chapter

The institute prescribes fifteen sections. Sections 1 to 5 are the title page, certificate, acknowledgement, declaration and index. Section 14 is the bibliography. The nine below are the ones your system supplies.

Chapter	What goes in it	Where it comes from
6. Executive Summary	Your question and your single headline finding	L3 result
7. Introduction to the Study	How this work is done manually today and what that costs	Desk research
8. Objectives of the Study	Your research question, stated so it can be answered either way	Section 6 of this document
9. Scope and Limitations	What you did not cover and where your data stops	assumptions.md
10. Research Methodology	The five layers, your data sources and your sampling choice	sources.csv
11. Data Analysis and Interpretation	Engine output, the validation test, and what it means. **This chapter carries 10 of the 20 report marks.**	L3 and the validation test
12. Findings and Discussion	Your answer, stated with its limits	L3 and L4
13. Conclusion and Suggestions	What a firm should do differently, and what it would cost	L5
15. Annexures	The generated report, screenshots of the running application, the repository link and sources.csv	L5 and the repository

> **Chapter 11 decides your report mark.** Half of the twenty report marks sit in data analysis and interpretation. Write it last, write it longest, and make sure every table in it is one you produced rather than one you copied.

5. How to Read Your Entry

Each entry in section 6 gives you eight things. Work through them in order.

Field	What to do with it
Project title	Your title, fixed. It goes on your title page exactly as printed.
System name	What to call your repository and your application.
Research question	Chapter 8 of your report. The answer must be able to come out either way.
Module artifacts	Which in-class outputs you reuse, so you do not start from an empty folder.
Secondary data source	Where your data comes from. Log every pull in sources.csv.
Engine	The calculation you must build and be able to explain. This is the twenty-mark layer.
Validation test	How you prove your engine is right. Put the result in chapter 11.
Viva demonstration	Your three minutes in front of the panel. Rehearse it.

Artifact codes: **A1** structured extraction (Module 1) · **A2** retrieval pipeline (Module 2) · **A3** agent pipeline (Module 3) · **A4** backtest harness (Module 4) · **A5** VaR and PD engine (Module 5) · **A6** regulatory reporting (Module 6) · **A7** governance note (Module 7) · **A8** fund administration engine (Module 8) · **A9** dashboard and commentary (Module 9).

6. Group-Wise Project Expectations

Eighty-seven groups. Your title is fixed and is printed against your names.

GROUP 1 | MODULE M8

AI Based Financial reporting to LPs and GPs

Nikhil Kumawat · Abhishek Shah

> **RESEARCH QUESTION**
> Can an LP capital account statement be reproduced from public AIF disclosures alone, and where does the reconstruction break?

SYSTEM NAME

LPReport — AIF Investor Reporting Engine

MODULE ARTIFACTS TO REUSE

A1, A8, A7, A9

SECONDARY DATA SOURCE

SEBI AIF disclosures; ILPA reporting template; listed PE/VC annual reports

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Capital account roll-forward: contributions, drawdowns, distributions, management fee accrual and carried interest under a European waterfall

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

The roll-forward closing balance ties to the fund NAV in the AIF disclosure, to the rupee

VIVA DEMONSTRATION, THREE MINUTES

Run the waterfall on one fund, show the tie-out, then break one input and show the exception fire

GROUP 2 | MODULE M5

Intelligent Portfolio Risk & Financial Insight Platform

Mimansa Priyadarsini · Sachin Bhadouria

> **RESEARCH QUESTION**
> How far does Monte Carlo VaR diverge from parametric VaR on NSE data, and what causes the gap?

SYSTEM NAME

RiskConsole — Portfolio VaR Engine

MODULE ARTIFACTS TO REUSE

A1, A5, A7, A9

SECONDARY DATA SOURCE

NSE bhavcopy; yfinance price history

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

10,000-path Monte Carlo on a Cholesky-decomposed covariance matrix; 95% and 99% VaR, CVaR, marginal VaR by holding

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Monte Carlo VaR converges on the parametric VaR for a normally distributed test portfolio, within a stated tolerance

VIVA DEMONSTRATION, THREE MINUTES

Run 10,000 paths live, show the convergence plot, name the holding contributing the most risk

GROUP 3 I MODULE M4

Statistical Learning-Based Quantitative Trading Strategy for the Indian Equity Market

Mahak Bansal · Ayush Narendra Kale

> **RESEARCH QUESTION**
> Does the in-sample edge survive walk-forward testing once costs are charged?

SYSTEM NAME

QuantLab — Walk-Forward Strategy Harness

MODULE ARTIFACTS TO REUSE

A1, A4, A7, A9

SECONDARY DATA SOURCE

NSE bhavcopy 2019-2026

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Price and volume feature set, walk-forward train-test split, position sizing, 20bps transaction cost model

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Out-of-sample Sharpe is reported separately from in-sample, on data the model never saw

VIVA DEMONSTRATION, THREE MINUTES

Show the in-sample and out-of-sample equity curves side by side and explain the gap

GROUP 4 I MODULE M6

BASEL RISK MIGRATION - AI-Powered Regulatory Arbitrage & Systemic Credit Intelligence

Tulshi Prasantbhai Pal · Naman Nihalani

> **RESEARCH QUESTION**
> How has RWA density moved across Indian banks since 2019, and what explains the divergence?

SYSTEM NAME

BaselTrack — Pillar 3 Comparison Engine

MODULE ARTIFACTS TO REUSE

A1, A2, A6, A7, A9

SECONDARY DATA SOURCE

RBI Pillar 3 disclosures for 10 banks; RBI DBIE

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

CET1, Tier 1, total capital, RWA density and leverage ratio recomputed from parsed Pillar 3 tables

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

The recomputed CET1 ratio matches the ratio the bank itself prints, for all ten banks

VIVA DEMONSTRATION, THREE MINUTES

Parse a Pillar 3 PDF live, show the recomputed ratio next to the printed one

GROUP 5 I MODULE M2

Advanced AI-Powered Tax Research

Abhijit Kumar Mandal · Pooja Chakraraj Joshi

RESEARCH QUESTION

How accurately does retrieval answer Income Tax Act questions, and which question types fail?

SYSTEM NAME

TaxRAG — Statutory Research Assistant

MODULE ARTIFACTS TO REUSE

A1, A2, A7, A9

SECONDARY DATA SOURCE

incometaxindia.gov.in Act text; CBDT circulars; ITAT orders

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Section-level chunking, embedding retrieval, re-ranking, and an answer that must cite the section it came from

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Twenty questions with known answers; precision@3 and citation accuracy are both reported

VIVA DEMONSTRATION, THREE MINUTES

Ask three questions including one it gets wrong, and show the sections it retrieved

GROUP 6 I MODULE M9

AI-Powered Equity Research and Valuation Assistant

Chopda Meet · Rupesh Kumar Ranjan

RESEARCH QUESTION

What does a DCF say about NSE IT services valuations relative to the traded price?

SYSTEM NAME

ITValuer — DCF and Comparables Engine

MODULE ARTIFACTS TO REUSE

A1, A2, A7, A9

SECONDARY DATA SOURCE

Annual reports; NSE filings; Damodaran datasets

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Discounted cash flow with a revenue build, WACC from CAPM and a terminal value, plus comparable company analysis on EV/EBITDA and P/E

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

The DCF output reconciles to a hand-built Excel model on identical inputs

VIVA DEMONSTRATION, THREE MINUTES

Move WACC by 100 basis points live and show the value swing

GROUP 7 I MODULE M5 / M8

AI-Based Mutual Fund Recommendation and Portfolio Optimization System

Prabhu Ashish Senapati · Aniket Dattaram Gawade

RESEARCH QUESTION

Do funds selected on rolling returns and style stability outperform a naive equal-weight allocation?

SYSTEM NAME

FundPicker — Selection and Allocation Engine

MODULE ARTIFACTS TO REUSE

A1, A5, A8, A7, A9

SECONDARY DATA SOURCE

AMFI NAV history; SEBI scheme information documents

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Rolling return and style drift screens, then mean-variance optimisation with a weight constraint set

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

The optimiser reproduces the textbook efficient frontier on a three-asset test case

VIVA DEMONSTRATION, THREE MINUTES

Run the screen, show the frontier, and show what changes when the constraint set is relaxed

GROUP 8 I MODULE M8 / M9

Rolling cash flow forecaster - Predicting 13-week or monthly cash flow using historical patterns + AR/AP aging data

Sakshi Kailas Sawant · Shrirang Vinod Hiwarkar

RESEARCH QUESTION

How far ahead can a 13-week cash forecast stay inside a 10% error band on ageing-driven inputs?

SYSTEM NAME

CashRoll — 13-Week Forecast Engine

MODULE ARTIFACTS TO REUSE

A1, A8, A7, A9

SECONDARY DATA SOURCE

Listed-company debtor and creditor ageing disclosures; a synthetic ledger calibrated to published DSO and DPO norms

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Collection curve fitted to ageing buckets, projected 13 weeks forward, with a weekly forecast-versus-actual variance loop

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Forecast error is measured against held-out weeks and reported as MAPE, not asserted

VIVA DEMONSTRATION, THREE MINUTES

Show week 1 to week 13 forecast against actual, and point at the week it breaks

GROUP 9 I MODULE M9

“AI-Driven FP&A Decision Intelligence: Automated Variance Analysis, Forecasting and Executive Reporting”

Poonam Mahendar Singh · HARESH SURESH PALVE

RESEARCH QUESTION

Which of price, volume or mix drove the margin change at this FMCG company over three years?

SYSTEM NAME

FPAEngine — Variance and Board Pack Generator

MODULE ARTIFACTS TO REUSE

A1, A7, A9

SECONDARY DATA SOURCE

Twelve quarters of results for one listed FMCG company

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Price, volume and mix decomposition of revenue variance, plus a margin bridge across twelve quarters

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

The decomposed variances sum back to the reported total variance, every quarter

VIVA DEMONSTRATION, THREE MINUTES

Generate the board pack live and read the auto-written commentary against the actual filing

GROUP 10 I MODULE M5

AI-Powered Robo-Advisory and Wealth Management

Choudhary Joniya Suresh Kumar · Chetana Adhiya

RESEARCH QUESTION

Do drift-band rebalancing rules beat calendar rebalancing after transaction costs?

SYSTEM NAME

RoboAdvise — Profiling and Rebalance Engine

MODULE ARTIFACTS TO REUSE

A1, A5, A7, A9

SECONDARY DATA SOURCE

AMFI NAV data; index history; SEBI Investment Adviser Regulations

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Scored risk questionnaire mapped to model portfolios, with rebalancing triggered by drift bands

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

The same investor profile always returns the same allocation, proved by a repeatability test

VIVA DEMONSTRATION, THREE MINUTES

Enter two investor profiles, show the two allocations, and run the drift simulation

GROUP 11 | MODULE M5 / M9

AI Working Capital & Cash Flow Early Warning System

Sapna Kumari · Japjee Ahuja

RESEARCH QUESTION

Does a cash conversion cycle signal flag mid-cap distress before the market prices it?

SYSTEM NAME

WCWatch — Working Capital Early Warning

MODULE ARTIFACTS TO REUSE

A1, A5, A7, A9

SECONDARY DATA SOURCE

Five years of annual report data for 50 mid-caps

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Cash conversion cycle, current ratio trend and interest coverage combined into a scored stress signal

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

The signal is backtested against firms that later reported distress, with hit rate and false positive rate stated

VIVA DEMONSTRATION, THREE MINUTES

Show the traffic-light board, then open one company the model flagged early

GROUP 12 | MODULE M3

FinAgent - Multi-Agent AI Equity Research & Portfolio Intelligence Platform

Sahba Kausar · Sumit Gorai

RESEARCH QUESTION

Does adding a reviewer agent measurably improve memo quality over a single-agent baseline?

SYSTEM NAME

FinAgent — Three-Agent Research Pipeline

MODULE ARTIFACTS TO REUSE

A1, A2, A3, A7, A9

SECONDARY DATA SOURCE

NSE announcements; bank annual reports; RBI data

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Research agent gathers, analysis agent computes bank metrics, writer agent drafts, reviewer agent scores the memo

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

The reviewer agent's score is checked against a human rubric on ten memos, and agreement is reported

VIVA DEMONSTRATION, THREE MINUTES

Run the pipeline live on one bank and show the reviewer agent rejecting a weak claim

GROUP 13 I MODULE M5

Smart Finance AI: Personal Debt and Repayment Optimization Platform

Vaishali Patel · Nishant Madan Ingole

> **RESEARCH QUESTION**
> Which repayment strategy saves the most interest, and how much does that answer change with the rate?

SYSTEM NAME

DebtOptim — Repayment Strategy Engine

MODULE ARTIFACTS TO REUSE

A1, A5, A7, A9

SECONDARY DATA SOURCE

RBI lending-rate data; published bank rate cards

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Amortisation schedules under avalanche, snowball and restructured EMI, with total interest and time-to-zero for each

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

The amortisation schedule ties to a bank's own published EMI figure for the same loan terms

VIVA DEMONSTRATION, THREE MINUTES

Load one debt stack, run all three strategies, and show the interest difference

GROUP 14 I MODULE M5 / M6

AI-Based Corporate Bond Credit Spread & Rating Migration Analyser

Gupta Sejal Rajkumar · Abhinav Panchal

> **RESEARCH QUESTION**
> Do credit spreads on Indian corporate bonds move before the rating agencies act?

SYSTEM NAME

SpreadWatch — Credit Spread Analyser

MODULE ARTIFACTS TO REUSE

A1, A2, A5, A7, A9

SECONDARY DATA SOURCE

NSE and BSE debt segment data; CRISIL and ICRA rating rationales; RBI data

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Yield spread over the matched-tenor government security, plus a rating transition matrix built from rating actions

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Computed spreads are checked against the traded yields printed in the exchange debt segment data

VIVA DEMONSTRATION, THREE MINUTES

Show the transition matrix and one issuer whose spread widened before the downgrade

GROUP 15 I MODULE M5

AI Portfolio & Risk Analyzer

Janhavi Jayram Gugale · Pranav Adhikar Mali

RESEARCH QUESTION

Did gold actually diversify an Indian equity portfolio in the drawdowns of the last ten years?

SYSTEM NAME

MultiAsset — Allocation and Drawdown Analyser

MODULE ARTIFACTS TO REUSE

A1, A5, A7, A9

SECONDARY DATA SOURCE

NSE indices; sovereign gold bond and gold prices; AMFI debt fund NAVs

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Rolling correlation across equity, gold and debt, plus maximum drawdown and recovery period per allocation mix

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Drawdown figures are reconciled against index drawdowns published for the same period

VIVA DEMONSTRATION, THREE MINUTES

Show the correlation heatmap, then overlay the drawdown of two allocation mixes

GROUP 16 I MODULE M8 / M9

PMS Services powered by AI

Satyam Mahadev Shelke · Kavi Katiyar

RESEARCH QUESTION

How much of PMS outperformance survives once fees are netted off?

SYSTEM NAME

PMSCompare — Strategy Disclosure Analyser

MODULE ARTIFACTS TO REUSE

A1, A8, A7, A9

SECONDARY DATA SOURCE

SEBI monthly PMS disclosure data

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Return, risk and fee normalisation across PMS strategies, with a net-of-fee ranking

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Recomputed returns match the returns SEBI publishes for each strategy

VIVA DEMONSTRATION, THREE MINUTES

Rank ten strategies gross, then net of fees, and show the order change

GROUP 17 I MODULE M1 / M9

An AI-powered platform for NSE/BSE stock analysis that extracts annual reports, performs fundamental and technical analysis, and generates buy/sell/hold signals to support smarter investment decisions

Dimple Chunnilal Pawar · Harsh Sarle

> **RESEARCH QUESTION**
> Does combining fundamental screens with technical signals beat either one alone on auto sector stocks?

SYSTEM NAME
AutoScreen — Extraction to Signal Pipeline

MODULE ARTIFACTS TO REUSE
A1, A2, A4, A7, A9

SECONDARY DATA SOURCE
Annual report PDFs; NSE price history

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND
Ratio extraction from annual report text, combined with RSI and moving average signals into a single scorecard

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT
Extracted line items are checked against the printed statements for 20 companies, with an accuracy percentage

VIVA DEMONSTRATION, THREE MINUTES
Extract from a PDF live, show the accuracy check, then show the combined signal

GROUP 18 I MODULE M9

Effective Tax Rates at Indian Listed Companies: Explaining the Gap from the Statutory Rate

Manish Tripathi · Rushikesh Mangesh Thorat

> **RESEARCH QUESTION**
> What explains the gap between the statutory and the effective tax rate at Indian listed companies?

SYSTEM NAME
TaxBridge — Effective Tax Rate Analyser

MODULE ARTIFACTS TO REUSE
A1, A2, A7, A9

SECONDARY DATA SOURCE
Annual report tax notes for 30 listed companies; CBDT rate notifications

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND
Statutory to effective tax rate reconciliation rebuilt from the tax note of 30 listed companies, decomposed into permanent differences, timing differences, incentives and prior-year adjustments

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT
Each rebuilt reconciliation ties to the tax expense the company reports, to the rupee

VIVA DEMONSTRATION, THREE MINUTES
Rebuild one company's tax bridge live and show it tying to the filed tax expense

GROUP 19 | MODULE M8

AI-Based Automation of NAV Calculation and Validation in Fund Accounting

Snehal Kailas Gade · John Martin

RESEARCH QUESTION

Where do NAV differences come from when a fund's NAV is rebuilt from its disclosed portfolio?

SYSTEM NAME

NAVEngine — Calculation and Exception Layer

MODULE ARTIFACTS TO REUSE

A1, A8, A7, A9

SECONDARY DATA SOURCE

AMFI published NAVs; monthly scheme portfolio disclosures

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Mark-to-market valuation, accrued expenses, units outstanding, NAV per unit, and a tolerance check against the published NAV

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Computed NAV matches the AMFI published NAV within the stated tolerance on every test day

VIVA DEMONSTRATION, THREE MINUTES

Compute the NAV for one date, show the tie to AMFI, then inject a stale price and show the exception

GROUP 20 | MODULE M5

Credit risk / loan default prediction model

Vedant Damodar Naik · Ankush Raj

RESEARCH QUESTION

Does the gradient boosted model beat logistic regression enough to justify losing interpretability?

SYSTEM NAME

PDScore — Default Prediction Scorecard

MODULE ARTIFACTS TO REUSE

A1, A5, A7, A9

SECONDARY DATA SOURCE

Public lending datasets (Lending Club, UCI German Credit); RBI sectoral credit data

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Logistic regression and XGBoost on financial ratio features, with Gini, KS and AUC-ROC, and SHAP for driver attribution

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Model performance is reported on a held-out test set, and the two models are compared on the same split

VIVA DEMONSTRATION, THREE MINUTES

Score one borrower live and show the SHAP waterfall explaining the decision

GROUP 21 | MODULE M9 / M5

Ai Driven Equity Research & Portfolio Management

Ayushi Jaiswal · Dependra Singh Sisodiya

RESEARCH QUESTION

Does a research-score-weighted pharma portfolio beat market-cap weighting?

SYSTEM NAME

PharmaAlloc — Research to Portfolio Engine

MODULE ARTIFACTS TO REUSE

A1, A2, A5, A7, A9

SECONDARY DATA SOURCE

Pharma sector annual reports; NSE price data

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Research scores converted to portfolio weights through a scoring-to-weight rule, with a concentration cap

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Weights sum to one and respect every stated cap, proved by a constraint test

VIVA DEMONSTRATION, THREE MINUTES

Generate one research note, show the score, and show the weight it produces

GROUP 22 | MODULE M4 / M5

AI Portfolio Rebalancing Manager

Soumya Ranjan Das · Jyoti

RESEARCH QUESTION

Which rebalancing rule delivers the best risk-adjusted return after turnover costs?

SYSTEM NAME

Rebalancer — Three-Rule Comparison Engine

MODULE ARTIFACTS TO REUSE

A1, A4, A5, A7, A9

SECONDARY DATA SOURCE

NSE index and stock history

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Calendar, threshold and volatility-targeted rebalancing, each backtested with turnover and cost tracking

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Turnover is computed and charged; results are reported before and after cost

VIVA DEMONSTRATION, THREE MINUTES

Run all three rules on the same portfolio and show the turnover difference

GROUP 23 I MODULE M9

AI - Assisted Financial Analysis & Dashboard Services For SMEs/Startups

Harsh Soni · Prachi Gupta

RESEARCH QUESTION

Which financial ratios separated the small caps that survived from those that did not?

SYSTEM NAME

SmallCapHealth — Ratio and Trend Dashboard

MODULE ARTIFACTS TO REUSE

A1, A7, A9

SECONDARY DATA SOURCE

MCA filings; small-cap annual reports

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Liquidity, leverage, profitability and efficiency ratios with three-year trend detection and outlier flagging

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Ratios are checked against the ratios the company itself discloses, where disclosed

VIVA DEMONSTRATION, THREE MINUTES

Open two companies side by side and show which ratio diverged first

GROUP 24 I MODULE M6 / M9

AI powered financial forensic analyst

Aditya Sandip Ghadwaje · Dhruv Sahu

RESEARCH QUESTION

Would the Beneish M-Score have flagged India's known accounting failures before enforcement action?

SYSTEM NAME

ForensiScan — Manipulation Screening Engine

MODULE ARTIFACTS TO REUSE

A1, A6, A7, A9

SECONDARY DATA SOURCE

Annual reports of companies subject to SEBI action, plus a matched control set

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Beneish M-Score with all eight indices, Altman Z-Score, and related-party transaction flagging

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

The screener is run on companies later subject to SEBI action and on a matched control set, with hit and false positive rates

VIVA DEMONSTRATION, THREE MINUTES

Score one enforcement case and one control company, and show the index that separated them

GROUP 25 | MODULE M4 / M5

Market Regime Detection and Allocation Switching: Does Regime-Aware Allocation Beat a Fixed Mix?

Piyush Agarwal · Joseph Josiah Cardozo

RESEARCH QUESTION

Does switching allocation by detected regime beat a fixed allocation after costs?

SYSTEM NAME

RegimeSwitch — Adaptive Allocation Engine

MODULE ARTIFACTS TO REUSE

A1, A4, A5, A7, A9

SECONDARY DATA SOURCE

NSE index history 2010-2026; India VIX

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Market regime detection from a volatility state or hidden Markov model, feeding a rule that switches allocation between equity and cash by regime

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Regime labels are inspected against known market events, and the switching rule is backtested out of sample against a fixed mix

VIVA DEMONSTRATION, THREE MINUTES

Show the regime timeline against the index, then the two equity curves side by side

GROUP 26 | MODULE M9 / M7

FinSmart AI: A Low-Cost AI-Enabled Financial Analytics Service for Small Businesses

Yogendra Singh Rana · Mahesh Sadashiv Bhupati

RESEARCH QUESTION

What does a small business give up by running financial analytics at zero software cost?

SYSTEM NAME

FinSmart — Zero-Cost Analytics Toolkit

MODULE ARTIFACTS TO REUSE

A1, A7, A9

SECONDARY DATA SOURCE

RBI and SIDBI MSME sector ratio data; listed micro-cap financials

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Ratio computation benchmarked against RBI and SIDBI MSME sector medians, plus a total cost of ownership model

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

The toolkit is run end to end with no paid API, and the TCO model is compared against a paid-stack equivalent

VIVA DEMONSTRATION, THREE MINUTES

Run the toolkit keyless, then show the TCO comparison against the paid stack

GROUP 27 I MODULE M6 / M9

Auditor Qualifications and Emphasis of Matter: What Do Modified Audit Opinions Predict?

Pranav Sandip Chavan · ANAMIKA SINGH

RESEARCH QUESTION

What do modified audit opinions predict about a company's subsequent performance?

SYSTEM NAME

AuditOpinion — Modified Opinion Study

MODULE ARTIFACTS TO REUSE

A1, A2, A7, A9

SECONDARY DATA SOURCE

Audit reports in annual reports of 100 NSE companies; NSE price data

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Audit opinions coded as clean, qualified, emphasis of matter or adverse, then tested against subsequent share price return, auditor change and restatement

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Coding is validated on a hand-checked sample and inter-coder agreement is reported

VIVA DEMONSTRATION, THREE MINUTES

Extract the opinion from one audit report live and show its coded category and outcome

GROUP 28 I MODULE M9

Which NSE Disclosure Categories Move the Price? Abnormal Returns by Announcement Type

Harshvardhan Soni · Utkarsha Bandhe

RESEARCH QUESTION

Which categories of NSE disclosure actually move the share price, and which are noise?

SYSTEM NAME

MaterialMove — Announcement Category Impact

MODULE ARTIFACTS TO REUSE

A1, A2, A7, A9

SECONDARY DATA SOURCE

NSE corporate announcement feed; NSE price data

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Announcements classified into categories, then abnormal returns measured against a market model, with significance testing per category

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

The estimation window is separated from the event window, and the test statistic is reported for every category

VIVA DEMONSTRATION, THREE MINUTES

Show the abnormal return by category chart, then route one live announcement to its category

GROUP 29 | MODULE M9

AI-Powered Equity Research & Investment Intelligence Platform

Rithika Shashidhar · Krishna Sanjay Bembade

RESEARCH QUESTION

Which bank metric best explained relative share price performance over the last three years?

SYSTEM NAME

BankResearch — Peer Metrics and Note Generator

MODULE ARTIFACTS TO REUSE

A1, A2, A7, A9

SECONDARY DATA SOURCE

Bank and NBFC annual reports; RBI data

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Net interest margin, gross and net NPA, provision coverage, CASA ratio and capital adequacy, compared across peers

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Every computed bank metric is checked against the figure disclosed in that bank's own results

VIVA DEMONSTRATION, THREE MINUTES

Compute the peer table live and show it matching the disclosed figures

GROUP 30 | MODULE M9

FINMATE AI — AI Virtual CFO for Small Businesses

RITIKA MAHURKAR · Rupali Shyam Mohol

RESEARCH QUESTION

What separates a small business that is funding-ready from one that is not, on public financials?

SYSTEM NAME

FinMate — Virtual CFO Assistant

MODULE ARTIFACTS TO REUSE

A1, A7, A9

SECONDARY DATA SOURCE

MSME sector benchmarks; listed micro-cap filings

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Ratio diagnostics, monthly cash runway, and a funding readiness score built from stated weighted criteria

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

The runway calculation is checked against a hand-worked example, and the score weights are documented

VIVA DEMONSTRATION, THREE MINUTES

Run the diagnostic on one company and show the readiness score decomposed

GROUP 31 | MODULE M9

"FINIOS" The Fundamental Analyst.

Harshal Rajendra Neve · Kushanki Kesharwani

> **RESEARCH QUESTION**
> Do quality screens identify consumer stocks that hold up better in drawdowns?

SYSTEM NAME

FINIOS — Fundamental Scoring Engine

MODULE ARTIFACTS TO REUSE

A1, A7, A9

SECONDARY DATA SOURCE

Consumer sector annual reports; NSE data

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Quality screens on return on capital, accruals and debt, feeding a composite score and a valuation band

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

The composite score is reproducible from published inputs, proved by rerunning it from raw data

VIVA DEMONSTRATION, THREE MINUTES

Score three consumer stocks and show the screen that separated them

GROUP 32 | MODULE M4 / M9

"MarketPulse India" — An AI-Powered Macro Intelligence & Multi-Factor Stock Recommendation Engine for the Indian Equity Market

Rohit Shrinivas Kore · Saahil Manoj Kingre

> **RESEARCH QUESTION**
> Do macro indicators predict which equity factor works next in India?

SYSTEM NAME

MarketPulse — Macro to Factor Screen

MODULE ARTIFACTS TO REUSE

A1, A4, A7, A9

SECONDARY DATA SOURCE

RBI DBIE; MOSPI releases; NSE data

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Macro indicators from RBI and MOSPI mapped to factor tilts, which then drive a stock screen

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Factor returns are computed cross-sectionally and tested for statistical significance

VIVA DEMONSTRATION, THREE MINUTES

Show the macro signal, the factor it selected, and the screen output for that month

GROUP 33 I MODULE M9

Do Management Guidance Ranges Predict Reported Results?

Sumit Kumar Mandal · Rabheru Harsh Manharlal

RESEARCH QUESTION

Do management guidance ranges predict reported results at Indian listed companies?

SYSTEM NAME

GuidanceTest — Forecast Accuracy Study

MODULE ARTIFACTS TO REUSE

A1, A2, A7, A9

SECONDARY DATA SOURCE

Quarterly filings and earnings call transcripts for 25 listed companies

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Management guidance ranges extracted from filings and earnings calls, compared against the figure actually reported, with hit rate and bias measured

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Every guidance-to-actual pair cites the filing it came from, so any pair can be checked

VIVA DEMONSTRATION, THREE MINUTES

Show the guidance versus actual scatter and open the filing behind the largest miss

GROUP 34 I MODULE M5 / M6

Bank Lending Concentration and Sectoral Credit Risk

Shubham Soni · Shivani Madanbhai Rathod

RESEARCH QUESTION

Does rising lending concentration in a sector precede rising stress in that sector?

SYSTEM NAME

CreditConc — Sectoral Lending Risk

MODULE ARTIFACTS TO REUSE

A1, A5, A7, A9

SECONDARY DATA SOURCE

RBI sectoral deployment of bank credit, monthly; RBI Financial Stability Report

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Sectoral credit concentration measured with a Herfindahl index, tracked against sectoral NPA and stress indicators over time

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Computed concentration is reconciled against the sectoral deployment totals RBI publishes

VIVA DEMONSTRATION, THREE MINUTES

Show the concentration index against sectoral NPA and name the sector with the widest gap

GROUP 35 | MODULE M8

use ML to forecast inflows/outflows for a category of funds based on market indicators, and discuss implications for liquidity management

Rishi Sharma · Abhishek Tiwari

RESEARCH QUESTION

Do market indicators predict mutual fund category flows well enough to plan liquidity?

SYSTEM NAME

FlowCast — Fund Flow Forecaster

MODULE ARTIFACTS TO REUSE

A1, A5, A8, A7, A9

SECONDARY DATA SOURCE

AMFI monthly category flow data; NSE index and interest rate data

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Regression of category net flows on market return, rate and volatility indicators, with a liquidity coverage calculation

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Forecast error is reported on held-out months, against a naive persistence benchmark

VIVA DEMONSTRATION, THREE MINUTES

Show the forecast against actual flows and the month it missed

GROUP 36 | MODULE M5

AI-Based Credit Risk Assessment and Loan Default Prediction

Karm Brijeshkumar Darji · Neha Pravin Kuralkar

RESEARCH QUESTION

Where should an MSME lender set its cut-off to balance approval rate against expected loss?

SYSTEM NAME

MSMEScore — Credit Scorecard and Cut-off

MODULE ARTIFACTS TO REUSE

A1, A5, A7, A9

SECONDARY DATA SOURCE

Public MSME lending data; RBI sectoral credit data

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Scorecard built from ratio features, plus a cut-off analysis trading approval rate against expected default rate

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

The cut-off curve is produced from the held-out set, and the chosen cut-off is justified in the report

VIVA DEMONSTRATION, THREE MINUTES

Move the cut-off live and show approval rate and expected loss move together

GROUP 37 I MODULE M4 / M5

AI-Based IPO Listing Gain Prediction Model for Indian IPOs

Shubham Digambar Sangle · Ashish Awasthi

RESEARCH QUESTION

Which IPO characteristic best predicts listing gains once subscription is controlled for?

SYSTEM NAME

IPOGain — Listing Return Model

MODULE ARTIFACTS TO REUSE

A1, A5, A7, A9

SECONDARY DATA SOURCE

NSE and BSE IPO data 2018-2026; DRHP issue details

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Regression of listing-day return on subscription multiples, issue size, grey market premium and sector

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

The model is tested out of sample on the twenty most recent IPOs, which were excluded from fitting

VIVA DEMONSTRATION, THREE MINUTES

Predict the last five IPOs live and show the prediction against what happened

GROUP 38 I MODULE M9

AI Based Stock Recommendation System with Buy , Sell and Hold Recommendation of that Stock with Reasons and Financial Data

Ayush Ghagre · Prajwal Rathore

RESEARCH QUESTION

Does a transparent weighted scorecard beat a coin flip on metals sector calls?

SYSTEM NAME

MetalScore — Transparent Rating Scorecard

MODULE ARTIFACTS TO REUSE

A1, A2, A7, A9

SECONDARY DATA SOURCE

Metals sector annual reports; NSE prices

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Weighted scorecard across valuation, quality and momentum, with the weights fixed and published in the report

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Rating hit rate is backtested over forward three-month returns, with the base rate stated for comparison

VIVA DEMONSTRATION, THREE MINUTES

Show the weights, score one stock, and show the backtested hit rate

GROUP 39 | MODULE M8

CollectAI: An AI-Powered Accounts Receivable Management and Payment Risk Prediction System for MSMEs

Anshuli Saxena · Sahu Premkumar

RESEARCH QUESTION

Which customer characteristics predict late payment in the published MSME delay data?

SYSTEM NAME

CollectAI — Receivables Risk Engine

MODULE ARTIFACTS TO REUSE

A1, A8, A7, A9

SECONDARY DATA SOURCE

MSME Samadhaan delayed-payment case data; listed company debtor ageing disclosures

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Ageing bucket analysis, days sales outstanding, and a delay risk score calibrated to published MSME payment delays

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

The risk score is calibrated against MSME Samadhaan outcome data, and calibration is plotted

VIVA DEMONSTRATION, THREE MINUTES

Score a receivables book and show the collection priority list it produces

GROUP 40 | MODULE M5

Personalized Stock Recommendation Engine Using Risk Profiling, Investment Horizon, and Capital Allocation

Bhojraj Krishna Kamdi · Vivek Rajesh Rudani

RESEARCH QUESTION

How much does position sizing, rather than stock selection, drive outcome dispersion?

SYSTEM NAME

SizeRight — Profile-Driven Allocation Engine

MODULE ARTIFACTS TO REUSE

A1, A5, A7, A9

SECONDARY DATA SOURCE

NSE data; SEBI investor risk-profiling guidance

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Risk tolerance and horizon mapped to position sizing through a volatility-scaled rule with a maximum position cap

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Position sizes are proved to respect the cap and to scale inversely with volatility, by test

VIVA DEMONSTRATION, THREE MINUTES

Enter two profiles and show the same stock sized differently for each

GROUP 41 | MODULE M1 / M2

AI Financial Document & Earnings Report Analyzer

Utkarsh Anand Pratap Singh · Priyanshu Jain

RESEARCH QUESTION

Does management language change before the reported numbers do?

SYSTEM NAME

EarningsRead — Filing Extraction Engine

MODULE ARTIFACTS TO REUSE

A1, A2, A7, A9

SECONDARY DATA SOURCE

NSE quarterly filings; earnings call transcripts

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Extraction of guidance statements, margin commentary and risk language, tracked as change across eight quarters

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Extraction accuracy is scored against a hand-labelled set, and reported as a percentage

VIVA DEMONSTRATION, THREE MINUTES

Extract from one quarterly filing live and show the change against the prior quarter

GROUP 42 | MODULE M4

Intelligent Stock Market Decision System Using AI

Tanishq Manoj Sapare · Kaustubh Narendra Boraste

RESEARCH QUESTION

Do combined technical rules improve hit rate, or only reduce the number of trades?

SYSTEM NAME

SignalDesk — Technical Rule Backtester

MODULE ARTIFACTS TO REUSE

A1, A4, A7, A9

SECONDARY DATA SOURCE

NSE price history

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

RSI, MACD and Bollinger rules coded separately and in combination, each backtested with hit rate and payoff ratio

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Every rule is tested on the same data window and the same cost assumption, so the comparison is fair

VIVA DEMONSTRATION, THREE MINUTES

Run all three rules on one stock and show the trade count and hit rate for each

GROUP 43 I MODULE M1 / M9

Cost Structure Analysis of Listed Companies: Where Operating Expense Ratios Diverge Within a Sector

Shravani Rajiv Pittulwar · Ragini Vinod Kumar Nishad

RESEARCH QUESTION

Where do operating expense ratios diverge within a sector, and does the divergence persist?

SYSTEM NAME

CostStruct — Operating Expense Analyser

MODULE ARTIFACTS TO REUSE

A1, A2, A7, A9

SECONDARY DATA SOURCE

Schedule III expense disclosures in annual reports across one NSE sector

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Schedule III expense heads normalised to revenue across a sector, with dispersion analysis and outlier identification by head

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Every extracted expense head ties to the printed statement, checked across the full sample

VIVA DEMONSTRATION, THREE MINUTES

Extract the expense heads from one annual report live and place the company in the sector spread

GROUP 44 I MODULE M6

AI-Powered Cooperative Banking Liquidity & NPA Early Warning Dashboard

Avkash Suresh Kothale · Amruta Pankaj Biswas

RESEARCH QUESTION

How many months before an RBI direction do published co-operative bank ratios turn?

SYSTEM NAME

UCBWatch — Co-operative Bank Early Warning

MODULE ARTIFACTS TO REUSE

A1, A5, A6, A7, A9

SECONDARY DATA SOURCE

RBI urban co-operative bank statistical tables; RBI Report on Trend and Progress of Banking

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Gross NPA trend, credit-deposit ratio, liquidity coverage and capital adequacy combined into a scored warning signal

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

The signal is backtested against urban co-operative banks RBI later placed under directions, with lead time reported

VIVA DEMONSTRATION, THREE MINUTES

Show the dashboard, then open a bank RBI later restricted and show the lead time

GROUP 45 | MODULE M9 / M4

Real-Time Financial News Summarization and Sentiment-Driven Trading Signal Generator

Vaibhav Hemraj Bramhankar · Shubham Patel

RESEARCH QUESTION

Does announcement sentiment carry tradable information after transaction costs?

SYSTEM NAME

SentiSignal — News to Trade Engine

MODULE ARTIFACTS TO REUSE

A1, A4, A7, A9

SECONDARY DATA SOURCE

NSE announcements; public news archives; NSE price data

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

FinBERT sentiment scoring on announcements, converted to an event-driven entry and exit rule with a holding period

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

The strategy is backtested with the sentiment score computed only from information available before the trade

VIVA DEMONSTRATION, THREE MINUTES

Score one announcement live, then show the backtested return of the rule it triggers

GROUP 46 | MODULE M4 / M5

AI WealthPilot - AI-Powered Investment Allocation, Trade Risk Assessment & Algorithmic Strategy Optimisation

Aayush Vishal Karwa · SOUMIK SADHUKHAN

RESEARCH QUESTION

How often do pre-trade risk limits bind, and what do they cost in returns?

SYSTEM NAME

WealthPilot — Allocation with Pre-Trade Risk

MODULE ARTIFACTS TO REUSE

A1, A4, A5, A7, A9

SECONDARY DATA SOURCE

NSE price and index data

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Allocation engine with pre-trade checks on concentration, liquidity and exposure limits, plus a parameter sweep

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Every blocked trade is logged with the limit it breached, and the log is shown as evidence

VIVA DEMONSTRATION, THREE MINUTES

Submit a trade that breaches a limit and show the block and the log entry

GROUP 47 | MODULE M2 / M9

Ai powered IPO Analyzer

Chirag Gupta · Harshal Saurkar

RESEARCH QUESTION

Do the risk factors disclosed in a DRHP predict post-listing underperformance?

SYSTEM NAME

DRHPReader — IPO Document Analyser

MODULE ARTIFACTS TO REUSE

A1, A2, A7, A9

SECONDARY DATA SOURCE

SEBI DRHP filings; NSE peer data

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Risk factor extraction and categorisation from the DRHP, plus peer multiple comparison at the issue price

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Extraction is scored against a hand-labelled risk factor set from three DRHPs

VIVA DEMONSTRATION, THREE MINUTES

Query one DRHP live and show the risk categories against the listing outcome

GROUP 48 | MODULE M1

Old versus New Tax Regime: Where the Choice Flips, and How Many Taxpayers Sit Near That Line

Vaidik Basal · Vedika Mahadev Belekar

RESEARCH QUESTION

At what income and deduction level does the regime choice flip, and how many taxpayers sit near that line?

SYSTEM NAME

RegimeLine — Old versus New Tax Regime

MODULE ARTIFACTS TO REUSE

A1, A2, A7, A9

SECONDARY DATA SOURCE

Income Tax Act provisions; CBDT rates; Income Tax Department taxpayer distribution statistics

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Both regimes computed across an income and deduction grid to locate the breakeven surface, overlaid on the published taxpayer income distribution

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Both regimes are validated against the worked examples the Income Tax Department publishes

VIVA DEMONSTRATION, THREE MINUTES

Show the breakeven chart with the taxpayer distribution overlaid, then compute one real case

GROUP 49 | MODULE M2 / M8

AI-Powered Mutual Fund Factsheet Summarizer + Red-Flag Detector

Saharsh Vinay Ganorkar · Sumit Suresh Shinde

> **RESEARCH QUESTION**
> How much portfolio overlap does an investor holding four large-cap funds actually have?

SYSTEM NAME

FactSheetAI — Overlap and Drift Detector

MODULE ARTIFACTS TO REUSE

A1, A2, A8, A7, A9

SECONDARY DATA SOURCE

AMFI factsheets; monthly portfolio disclosures

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Portfolio overlap percentage between schemes, top-ten concentration, and style drift measured against the stated mandate

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Overlap is recomputed from the disclosed monthly portfolios and cross-checked between two schemes

VIVA DEMONSTRATION, THREE MINUTES

Compare two schemes live and show the overlap percentage and the drifted holdings

GROUP 50 | MODULE M3 / M5

AI Investment Research & Portfolio Risk Platform

Kanishka Parihar · Priyansh Sangotra

> **RESEARCH QUESTION**
> Do the highest-conviction research calls also carry the highest risk contribution?

SYSTEM NAME

ResearchToRisk — Linked Pipeline

MODULE ARTIFACTS TO REUSE

A1, A3, A5, A7, A9

SECONDARY DATA SOURCE

NSE data; annual reports

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Research conclusions converted to weights, then run through a risk contribution calculation by holding

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Risk contributions sum to total portfolio risk, proved arithmetically in the report

VIVA DEMONSTRATION, THREE MINUTES

Show the memo, the weight it produced, and that holding's share of portfolio risk

GROUP 51 | MODULE M5

AI Loan & Credit Assessment Engine

Deepali Agrawal · Achal Ghumre

RESEARCH QUESTION

How much does risk-based pricing change approval economics against flat pricing?

SYSTEM NAME

LoanDesk — Eligibility and Pricing Engine

MODULE ARTIFACTS TO REUSE

A1, A5, A7, A9

SECONDARY DATA SOURCE

RBI lending rate data; published bank credit policies

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Eligibility rules, fixed obligation to income ratio affordability test, and risk-based pricing over a base rate

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Every decision writes an audit log naming the rule that decided it

VIVA DEMONSTRATION, THREE MINUTES

Run one application through, show the decision log, then change the income and rerun

GROUP 52 | MODULE M8

Fund Expense Allocation & Accrual Automation Model

Nancy Keshri · Prachi Gajanan Barsagade

RESEARCH QUESTION

Where do funds sit against their SEBI TER caps, and how much does that cost investors annually?

SYSTEM NAME

AccrualEngine — TER and Expense Allocation

MODULE ARTIFACTS TO REUSE

A1, A8, A7, A9

SECONDARY DATA SOURCE

SEBI TER regulations; AMFI scheme documents

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Daily expense accrual, allocation across plans, TER computation and a check against the SEBI slab cap

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

The computed TER is checked against the TER the scheme itself discloses

VIVA DEMONSTRATION, THREE MINUTES

Accrue one day live, show the TER, and show the cap check firing on a breach

GROUP 53 I MODULE M4

AlphaGuard AI: Multi-Asset Algorithmic Trading & Risk-Managed Investment Advisory System

Takvo Topden Lepcha · Jeet Subhash Thavrani

> **RESEARCH QUESTION**
> Does a drawdown circuit breaker improve risk-adjusted return, or only reduce return?

SYSTEM NAME

AlphaGuard — Multi-Asset Risk Overlay

MODULE ARTIFACTS TO REUSE

A1, A4, A5, A7, A9

SECONDARY DATA SOURCE

NSE equity, gold and index data

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Entry signals per asset, volatility-scaled position sizing, stop rules and a portfolio drawdown circuit breaker

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

The circuit breaker is proved to fire in backtest at the stated drawdown level, every time

VIVA DEMONSTRATION, THREE MINUTES

Run the backtest to the drawdown limit and show the breaker cutting exposure

GROUP 54 I MODULE M9

AI IPO Research & Valuation Assistant

Aditi Yadnyadatt Pingle · Ananya Anand

> **RESEARCH QUESTION**
> Are IPOs priced above peer multiples systematically worse investments after listing?

SYSTEM NAME

IPOValue — Pricing Band Assessment

MODULE ARTIFACTS TO REUSE

A1, A2, A7, A9

SECONDARY DATA SOURCE

DRHP filings; NSE listing and post-listing price data

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Issue price multiples compared against listed peer multiples, tracked against post-listing returns at 1, 3 and 6 months

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Peer multiples are recomputed from filings rather than taken from a data vendor

VIVA DEMONSTRATION, THREE MINUTES

Show one IPO's premium to peers and its six-month return

GROUP 55 I MODULE M9

AI MIS & Dashboard Automation for SMEs Project

Mehak Sachin Singhal · Yash Agrawal

RESEARCH QUESTION

How much management reporting can be automated straight from public filings?

SYSTEM NAME

MISAuto — Filing to Dashboard Pipeline

MODULE ARTIFACTS TO REUSE

A1, A7, A9

SECONDARY DATA SOURCE

Quarterly filings of one listed company

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Quarterly filing parsed into a monthly management view with variance against prior period and auto-written commentary

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

The parsed figures tie to the filed statement line by line

VIVA DEMONSTRATION, THREE MINUTES

Load a filing, show the dashboard rebuild, and read the generated commentary

GROUP 56 I MODULE M9

AI-Powered Management Commentary Analyzer for Financial Decision-Making

Khushi Arun Shende · Akshat Jain

RESEARCH QUESTION

Does management tone in one quarter predict the results of the next?

SYSTEM NAME

ToneIndex — Management Commentary Analyser

MODULE ARTIFACTS TO REUSE

A1, A2, A7, A9

SECONDARY DATA SOURCE

MD&A sections; earnings call transcripts

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Tone scoring, hedging word frequency and guidance change detection across twelve quarters

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

The tone index is compared against subsequent reported results, and the correlation is stated

VIVA DEMONSTRATION, THREE MINUTES

Show the tone timeline against the earnings surprises

GROUP 57 | MODULE M9

Stock Market Sentiment Analysis

Anubhav Sanjay Thombare · Radhika Gupta

RESEARCH QUESTION

Does FinBERT sentiment on NSE news predict next-day abnormal returns at a significant level?

SYSTEM NAME

SentiTest — Event Study Engine

MODULE ARTIFACTS TO REUSE

A1, A7, A9

SECONDARY DATA SOURCE

NSE announcements; news archives; price data

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

FinBERT sentiment scoring, abnormal return computation against a market model, and a t-test on the mean

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

The event study uses a clean estimation window and reports statistical significance, not just direction

VIVA DEMONSTRATION, THREE MINUTES

Show the cumulative abnormal return plot and the p-value

GROUP 58 | MODULE M9

AI powered merger and acquisition target screening & valuation model

Shreyas Sameer Pujari · Gaurav Subhash Hajare

RESEARCH QUESTION

What premium have Indian mid-cap acquirers actually paid, and what explains the spread?

SYSTEM NAME

DealScreen — M&A Target and Premium Model

MODULE ARTIFACTS TO REUSE

A1, A2, A7, A9

SECONDARY DATA SOURCE

Completed Indian M&A deal disclosures; annual reports

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Target screening on size, growth and leverage fit, valuation range by multiple, and premium analysis against completed deals

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

The premium study uses only completed deals with disclosed consideration

VIVA DEMONSTRATION, THREE MINUTES

Screen for targets live, then show the premium distribution from completed deals

GROUP 59 | MODULE M8

AI-Based Corporate Actions Processing & Reconciliation Engine

Khushbu Prashant Jaiswal · Sachin Sudhir Kanojiya

> **RESEARCH QUESTION**
> Which corporate action types cause the most reconciliation breaks, and why?

SYSTEM NAME

CorpActions — Processing and Reconciliation Engine

MODULE ARTIFACTS TO REUSE

A1, A8, A7, A9

SECONDARY DATA SOURCE

NSE corporate action files; AMFI scheme portfolios

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Dividend, bonus, split and rights processing with holdings adjustment, ex-date handling and a reconciliation break report

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Adjusted holdings reconcile to the post-action position, and every break is explained

VIVA DEMONSTRATION, THREE MINUTES

Process a bonus and a split, then inject a wrong ex-date and show the break

GROUP 60 | MODULE M3

"AI Deal Team: An Agentic Pipeline for End-to-End M&A Deal Valuation & Structuring

Amit Anandrao Patil · Trupti Vikas Tekawade

> **RESEARCH QUESTION**
> Where does a multi-agent deal pipeline fail, and which agent causes most of the errors?

SYSTEM NAME

DealTeam — Four-Agent M&A Pipeline

MODULE ARTIFACTS TO REUSE

A1, A2, A3, A7, A9

SECONDARY DATA SOURCE

Completed deal disclosures; annual reports

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Screening agent, valuation agent, synergy agent and structuring agent, each with a defined tool set and handoff

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Each agent's output is checked independently, so a failure can be traced to the agent that caused it

VIVA DEMONSTRATION, THREE MINUTES

Run the pipeline and show the handoff trace, including one agent's rejected output

GROUP 61 | MODULE M5

AI-Based Prediction of Stock Volatility Risk Using Financial and Market Indicators: An Empirical Study of Indian Listed Companies

Vipul Kumar Singh · Shruti Sanjay Paradkar

RESEARCH QUESTION

Does a machine learning model beat GARCH at forecasting Indian equity volatility?

SYSTEM NAME

VolForecast — GARCH and ML Comparison

MODULE ARTIFACTS TO REUSE

A1, A5, A7, A9

SECONDARY DATA SOURCE

NSE price history; annual report ratios

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

GARCH(1,1) volatility forecast benchmarked against a machine learning forecast on identical features

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Forecast error is compared on the same held-out window using the same loss function

VIVA DEMONSTRATION, THREE MINUTES

Show both forecasts against realised volatility and state which won

GROUP 62 | MODULE M9

Does Announcement Timing Affect Price Reaction? Pre-Market versus Post-Market NSE Filings

Sudhanshu Kumar · Abhishek Mahesh Chavan

RESEARCH QUESTION

Does the time of day a filing is made change how the market reacts to it?

SYSTEM NAME

TimingEffect — Filing Time and Price Reaction

MODULE ARTIFACTS TO REUSE

A1, A2, A7, A9

SECONDARY DATA SOURCE

NSE filing timestamps; NSE price and volume data

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Filings bucketed by timestamp into market hours, post-market and pre-market, with abnormal return and volume measured per bucket

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Timestamps are taken from the exchange filing record itself, and the bucketing rule is stated in full

VIVA DEMONSTRATION, THREE MINUTES

Show abnormal return by timing bucket and open one after-hours filing and its next-day move

GROUP 63 | MODULE M1 / M9

AI Based Financial Statement Analyzer

Vaishnavi Kundalik Shinde · Gourang Gurudas Haldar

RESEARCH QUESTION

How accurately can annual report line items be extracted without manual entry?

SYSTEM NAME

StatementAI — Extraction and Trend Engine

MODULE ARTIFACTS TO REUSE

A1, A2, A7, A9

SECONDARY DATA SOURCE

Cement sector annual reports

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Line item extraction from annual reports, ratio computation and three-year trend commentary

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Extraction accuracy is measured line by line against the printed statements for the whole sector

VIVA DEMONSTRATION, THREE MINUTES

Extract one report live and show the accuracy table

GROUP 64 | MODULE M7

AI Governance & Model Risk Framework for Financial LLM Tools

Bhushan Nitin Khamankar · Sahil Ravindra Gawande

RESEARCH QUESTION

How easily can a finance LLM tool be made to leak or misstate, and which controls stop it?

SYSTEM NAME

RedTeamFin — LLM Risk and Governance Harness

MODULE ARTIFACTS TO REUSE

A1, A2, A7, A9

SECONDARY DATA SOURCE

OWASP LLM Top 10; Federal Reserve SR 11-7; EU AI Act; DPDP Act 2023

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Prompt injection test suite against a finance chatbot, scored by attack success rate, plus an SR 11-7 governance blueprint

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Every attack is logged with its payload and outcome, so the success rate is reproducible

VIVA DEMONSTRATION, THREE MINUTES

Run a live injection that succeeds, apply the control, and run it again

GROUP 65 | MODULE M2 / M9

AI BASED ESG RISK SCORING OF COMPANIES

Piyush Rohit Darji · Abhay Pratap Singh Rajput

RESEARCH QUESTION

Where do BRSR-derived scores disagree with commercial ESG ratings, and why?

SYSTEM NAME

ESGScore — BRSR Extraction and Scoring

MODULE ARTIFACTS TO REUSE

A1, A2, A7, A9

SECONDARY DATA SOURCE

SEBI BRSR filings

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

BRSR field extraction across nine principles, normalised and weighted into a transparent score

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

The score is compared against commercial ESG ratings, and disagreements are examined rather than hidden

VIVA DEMONSTRATION, THREE MINUTES

Score one company from its BRSR and show the disagreement with the commercial rating

GROUP 66 | MODULE M9 / M4

AI-Based Stock Market Event Impact Analyzer

Pranil Prakash Amrutkar · Pranay Dadghaye

RESEARCH QUESTION

Which announcement type moves NSE prices most, once the market model is controlled for?

SYSTEM NAME

EventImpact — Abnormal Return Engine

MODULE ARTIFACTS TO REUSE

A1, A4, A7, A9

SECONDARY DATA SOURCE

NSE announcement and price data

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Market model estimation, abnormal and cumulative abnormal returns around three event types, with significance testing

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Estimation and event windows are stated and separated, and the test statistic is reported

VIVA DEMONSTRATION, THREE MINUTES

Show the CAR plot for all three event types on one chart

GROUP 67 | MODULE M5 / M9

AI-powered Investment monitoring and opportunity discovery platform.

Atharv Dhananjay Yawale · Harshali Dinesh Shinde

RESEARCH QUESTION

How often does a portfolio drift past its bands in a year, and what does ignoring it cost?

SYSTEM NAME

PortMonitor — Drift Alerts and Screener

MODULE ARTIFACTS TO REUSE

A1, A5, A7, A9

SECONDARY DATA SOURCE

NSE price and corporate data

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Holdings drift measured against target weights, with promotion rules moving a screened stock onto the watchlist

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Every alert writes a log entry naming the rule and the threshold that fired it

VIVA DEMONSTRATION, THREE MINUTES

Show the drift chart, then trigger an alert and open its log entry

GROUP 68 | MODULE M2 / M3

Agentic AI for M&A Deal Evaluation: Financial Due Diligence, Valuation, Synergy Analysis and Risk Assessment

Sakshi Kiran Dhoot · Avantika Kishor Chavan

RESEARCH QUESTION

How much of a financial due diligence checklist can be answered from public documents alone?

SYSTEM NAME

DDAgent — Document-Grounded Diligence

MODULE ARTIFACTS TO REUSE

A1, A2, A3, A7, A9

SECONDARY DATA SOURCE

Annual reports and deal disclosures for one completed deal

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Due diligence checklist executed against indexed deal documents, with every answer carrying a document citation

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Answers without a retrievable citation are rejected by the system and counted, giving a refusal rate

VIVA DEMONSTRATION, THREE MINUTES

Run the checklist and show one item the system refused to answer for lack of evidence

GROUP 69 I MODULE M5

AI- Powered Personal Finance Manager

Nishant Vijay Rindhe · Rohit Vijay Tirlotkar

RESEARCH QUESTION

How much does asset placement between taxable and tax-advantaged accounts change a goal outcome?

SYSTEM NAME

GoalPlan — Personal Finance Engine

MODULE ARTIFACTS TO REUSE

A1, A5, A7, A9

SECONDARY DATA SOURCE

AMFI NAV data; income tax rules; CPI inflation data

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Goal corpus computation with inflation, SIP sizing from the future value formula, and tax-efficient asset placement

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

SIP figures are validated against a published SIP calculator on identical assumptions

VIVA DEMONSTRATION, THREE MINUTES

Enter one goal, show the SIP, then move the assets and show the after-tax difference

GROUP 70 I MODULE M9

AI-Powered Equity Performance Attribution Analysis in Indian Equity Mutual Funds

Nalin Jain · Somya Gupta

RESEARCH QUESTION

Did the outperformance or underperformance of selected Indian large-cap equity funds against their benchmark come from sector allocation or from stock selection?

SYSTEM NAME

AttribIQ — Equity Fund Performance Attribution Engine

MODULE ARTIFACTS TO REUSE

A1, A5, A7, A9

SECONDARY DATA SOURCE

AMC monthly portfolio disclosures (SEBI-mandated); AMFI NAV history; NSE index factsheets for benchmark sector weights; NSE sectoral index and closing price data

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Brinson-Fachler attribution of a fund's monthly active return against its benchmark, split by sector into allocation, selection and interaction effects, then linked across twelve months

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

For every month the three effects add exactly to the holdings-based active return, and the gap between the holdings-based return and the published NAV return is reported and explained

VIVA DEMONSTRATION, THREE MINUTES

Pick one fund and one month, show the allocation, selection and interaction bars adding to the active return, then load a month with a missing holding and show the engine refuse it

GROUP 71 | MODULE M9

AI-Powered Financial Decision Intelligence & Scenario Simulator for Corporate Management

Ashutosh Paul · Sachin Dilip Singh

RESEARCH QUESTION

Which shock breaks this company's covenants first, and at what magnitude?

SYSTEM NAME

ScenarioSim — Three-Statement Shock Model

MODULE ARTIFACTS TO REUSE

A1, A7, A9

SECONDARY DATA SOURCE

Annual report or 10-K of one listed company

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Linked income statement, balance sheet and cash flow, shocked on revenue, margin and capex, with the balance sheet forced to balance

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

The model balances in every scenario; a non-balancing scenario is treated as a failed test

VIVA DEMONSTRATION, THREE MINUTES

Run a 20% revenue shock live and show the balance sheet still balancing

GROUP 72 | MODULE M5 / M9

Transmission of US Fed Rate Changes to Indian Equity and Bond Markets: An Event Study

Tanaya Gajendra Khapre · Poonam Anil Hajare

RESEARCH QUESTION

How does a US Fed rate change transmit to Indian equity and bond markets, and how fast?

SYSTEM NAME

FedTransmit — Policy Rate Transmission Study

MODULE ARTIFACTS TO REUSE

A1, A5, A7, A9

SECONDARY DATA SOURCE

FRED policy rate series; RBI G-Sec yields; NSE index data

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Event study around FOMC rate decisions, measuring abnormal moves in NSE indices, the rupee and Indian government bond yields

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Estimation and event windows are stated and separated, and significance is tested rather than asserted

VIVA DEMONSTRATION, THREE MINUTES

Show the cumulative abnormal move across all three markets around one FOMC date

GROUP 73 I MODULE M9

AI CFO — Financial Decision Support Assistant

Rahul Shivprasad Pal · Gauri Jaydev Atkari

RESEARCH QUESTION

Does this mid-cap's actual funding mix minimise its cost of capital?

SYSTEM NAME

CFOAssist — Corporate Decision Models

MODULE ARTIFACTS TO REUSE

A1, A7, A9

SECONDARY DATA SOURCE

Annual reports of one listed mid-cap

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Working capital requirement, funding mix with a weighted average cost of capital calculation, and a dividend payout model

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

The WACC computed by the tool matches a hand-worked calculation on the same capital structure

VIVA DEMONSTRATION, THREE MINUTES

Change the debt-equity mix live and show WACC move against the actual mix

GROUP 74 I MODULE M1 / M9

AI-Driven Model for Financial Statement and Ratio Analysis

Dnyanesh Arvind Bhosle · Arpita Ravindra Pawar

RESEARCH QUESTION

What accuracy rate does automated extraction achieve across fifty annual reports, and where does it fail?

SYSTEM NAME

RatioScale — Extraction Pipeline at Scale

MODULE ARTIFACTS TO REUSE

A1, A2, A7, A9

SECONDARY DATA SOURCE

Annual reports of 50 listed companies

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Automated extraction across fifty companies with peer benchmarking by sector and an accuracy audit layer

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

A random sample of extracted values is manually checked, and the audited accuracy rate is published

VIVA DEMONSTRATION, THREE MINUTES

Show the audit sample, the accuracy rate, and one document that broke the parser

GROUP 75 I MODULE M9

“AI-Based Banking Sector Stock Analyzer: A Buy, Hold, or Sell Recommendation System”

Niranjana Chanda · Aniket Gupta

> **RESEARCH QUESTION**
> Which bank-specific metric drives the rating most, and does the rating track subsequent returns?

SYSTEM NAME

BankRate — Sector Analyser and Rating

MODULE ARTIFACTS TO REUSE

A1, A7, A9

SECONDARY DATA SOURCE

Bank annual reports; RBI data

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Asset quality, net interest margin, provision coverage and capital metrics feeding a transparent weighted rating

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Every metric is reconciled to the bank's disclosed figure before it enters the rating

VIVA DEMONSTRATION, THREE MINUTES

Rate three banks and show the metric that separated the top from the bottom

GROUP 76 I MODULE M4 / M9

An AI-Based Framework for Forecasting Stock Returns Using Corporate Financial Statements

ANISHA SARRAF · Adarsh Choudhury

> **RESEARCH QUESTION**
> Which fundamental factor earned a significant return premium on NSE data?

SYSTEM NAME

FactorLab — Fundamental Factor Study

MODULE ARTIFACTS TO REUSE

A1, A4, A7, A9

SECONDARY DATA SOURCE

Annual report data; NSE return data

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Factor construction from statement data, cross-sectional regression of forward returns, with significance testing

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Factors are constructed with a reporting lag so no future information leaks into the test

VIVA DEMONSTRATION, THREE MINUTES

Show the factor return series and the t-statistic for each factor

GROUP 77 I MODULE M6 / M9

Related Party Transactions: Disclosure Quality and Governance Outcomes

Aditya Adhikrao Mohite · Shroff Anurag Raju

RESEARCH QUESTION

Does related party transaction intensity relate to governance outcomes and subsequent performance?

SYSTEM NAME

RPTScan — Related Party Disclosure Study

MODULE ARTIFACTS TO REUSE

A1, A2, A6, A7, A9

SECONDARY DATA SOURCE

Related party transaction notes under SEBI LODR for 100 NSE companies

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Related party transactions extracted and scaled to revenue and to net worth, scored for disclosure quality against the LODR requirements

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Every extracted transaction cites the note it came from, and totals tie to the disclosed aggregate

VIVA DEMONSTRATION, THREE MINUTES

Extract the RPT note from one annual report live and show the company against its sector

GROUP 78 I MODULE M4 / M9

Do Sector Rotation Signals Work in India? Relative Strength Across NSE Sector Indices

Nikhil Dinesh Chauhan · Bhoomika B M

RESEARCH QUESTION

Do sector rotation signals beat holding the index in India, once turnover is charged?

SYSTEM NAME

SectorRotate — Relative Strength Study

MODULE ARTIFACTS TO REUSE

A1, A4, A7, A9

SECONDARY DATA SOURCE

NSE sector index history 2012-2026

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Relative strength ranking across NSE sector indices, with a rotation rule holding the top-ranked sectors, backtested against a buy-and-hold index

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Ranking uses only data available at the rebalance date, and costs are charged at every rotation

VIVA DEMONSTRATION, THREE MINUTES

Show the rotation equity curve against the index and the sectors it held through one drawdown

GROUP 79 | MODULE M9

AI-Based Detection of Overvalued and Undervalued Companies

Makhija Bhupendrasing Gulabsing · Aditi Anant Sutar

RESEARCH QUESTION

Do stocks screened as undervalued actually deliver higher forward returns?

SYSTEM NAME

MispriceScan — Relative Valuation Screen

MODULE ARTIFACTS TO REUSE

A1, A7, A9

SECONDARY DATA SOURCE

Annual reports; NSE price history

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Relative valuation screens on P/E, EV/EBITDA and P/B against sector medians, tested against forward returns

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Forward returns are measured from the screen date onward, with no look-ahead in the screen itself

VIVA DEMONSTRATION, THREE MINUTES

Show the screen output from two years ago and what those stocks did afterwards

GROUP 80 | MODULE M5 / M6

AI Powered Credit Rating Analysis

Sudhanshu Prashant Pande · Pratik Ravi Solanki

RESEARCH QUESTION

How much of a CRISIL or ICRA rating is explained by public financial ratios alone?

SYSTEM NAME

RatingReplica — Agency Decision Model

MODULE ARTIFACTS TO REUSE

A1, A2, A5, A7, A9

SECONDARY DATA SOURCE

CRISIL and ICRA rating rationales; annual reports

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Ordered model mapping financial ratios to rating notches, with an agreement matrix against the published rating

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Agreement is reported as exact-notch and within-one-notch accuracy on a held-out set

VIVA DEMONSTRATION, THREE MINUTES

Predict a rating live and show it against the agency's actual rating and rationale

GROUP 81 | MODULE M2 / M8

AI-Powered Private Equity Fund Intelligence & Investor Reporting System Using RAG

Waghmare Onkar Arun · Sahil Badrinath Basvekar

> **RESEARCH QUESTION**
> How much of an LP's quarterly question set can be answered from fund documents automatically?

SYSTEM NAME

PEReport — Fund Document Assistant

MODULE ARTIFACTS TO REUSE

A1, A2, A8, A7, A9

SECONDARY DATA SOURCE

SEBI AIF disclosures; listed PE firm reports; ILPA templates

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Retrieval over fund documents for investor queries, plus a drafted quarterly report assembled from retrieved facts

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Answer accuracy is scored on a fixed question set, and unsupported answers are refused and counted

VIVA DEMONSTRATION, THREE MINUTES

Ask three investor questions and show one refusal where the document did not support an answer

GROUP 82 | MODULE M5

AI-Powered Company financial health/credit risk screener

NIMISHA SINGH · Diksha Satish Gaonkar

> **RESEARCH QUESTION**
> Does the Altman Z-Score still separate distressed Indian companies from healthy ones?

SYSTEM NAME

HealthScreen — Distress Ranking Engine

MODULE ARTIFACTS TO REUSE

A1, A5, A7, A9

SECONDARY DATA SOURCE

Annual reports of 100 listed companies

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Altman Z-Score, interest coverage and leverage trend combined into a distress ranking across a hundred companies

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

The ranking is validated against companies that actually defaulted, with the hit rate at each decile

VIVA DEMONSTRATION, THREE MINUTES

Show the decile chart with actual defaults marked on it

GROUP 83 I MODULE M5 / M8

AI Treasury Risk & Liquidity Optimizer

Ayush Bhardwaj · Nakul Singh Bhadouriya

RESEARCH QUESTION

How much yield does a corporate treasury give up for each week of extra liquidity?

SYSTEM NAME

TreasuryOpt — Liquidity and Ladder Model

MODULE ARTIFACTS TO REUSE

A1, A5, A8, A7, A9

SECONDARY DATA SOURCE

RBI money market rates; T-bill auction data; company cash and investment disclosures

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Minimum cash buffer sizing, a short-term investment maturity ladder, and rate sensitivity of the ladder

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Ladder yields are checked against actual T-bill auction cut-offs for the same tenors

VIVA DEMONSTRATION, THREE MINUTES

Build the ladder live, then shorten it and show the yield given up

GROUP 84 I MODULE M2 / M9

Can AI Detect “Greenwashing” in Corporate Financial Reports?

Trupati Vilas Narkhede · Tanmay Pravin Dhonde

RESEARCH QUESTION

How often do stated ESG claims contradict the metrics reported alongside them?

SYSTEM NAME

GreenCheck — Claim Consistency Engine

MODULE ARTIFACTS TO REUSE

A1, A2, A7, A9

SECONDARY DATA SOURCE

SEBI BRSR filings; company sustainability reports

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

ESG claim extraction from narrative text, matched against the quantitative metric reported in the same BRSR

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Every flagged inconsistency cites both the claim text and the contradicting metric, so it can be verified

VIVA DEMONSTRATION, THREE MINUTES

Show one flagged company with the claim and the number side by side

GROUP 85 | MODULE M2 / M6

AI-Powered Green Bond Risk & Credibility Analyzer

Nandini Sanjay Kapse · Jitu Vyankatesh Annewar

RESEARCH QUESTION

Which of the four Green Bond Principles do Indian issuers report against least?

SYSTEM NAME

GreenBondIQ — Principles Compliance Scorer

MODULE ARTIFACTS TO REUSE

A1, A2, A6, A7, A9

SECONDARY DATA SOURCE

SEBI green bond filings; ICMA Green Bond Principles; issuer impact reports

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Use of proceeds, project evaluation, proceeds management and reporting scored against the four ICMA Green Bond Principles

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Each score cites the document paragraph that supports it, so the panel can check any score

VIVA DEMONSTRATION, THREE MINUTES

Score one issuance principle by principle and open the cited paragraph

GROUP 86 | MODULE M9

AI-Powered Voice of Customer Analytics for Financial Product and Service Improvement

Kunal Tiwari · Nayan Kumar Hota

RESEARCH QUESTION

Which product defects generate the most complaints across Indian financial products?

SYSTEM NAME

VoCFinance — Complaint and Review Mining

MODULE ARTIFACTS TO REUSE

A1, A2, A7, A9

SECONDARY DATA SOURCE

RBI Ombudsman reports; SEBI SCORES; public app store review data

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Topic modelling over complaint and review text, mapped to a product defect taxonomy with volume trends

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

Topic labels are validated against a hand-coded sample, and inter-rater agreement is reported

VIVA DEMONSTRATION, THREE MINUTES

Show the topic model output and the defect taxonomy it mapped to

GROUP 87 I MODULE M2 / M7

AI-Powered Investment Verification & Due Diligence: Evidence Behind Every Investment Decision.

Rashmi Sanjay Chate · Japleen Kaur

RESEARCH QUESTION

How much does a mandatory citation layer reduce hallucination in financial research output?

SYSTEM NAME

VerifyLayer — Citation-Grounded Verification

MODULE ARTIFACTS TO REUSE

A1, A2, A7, A9

SECONDARY DATA SOURCE

Annual reports; NSE filings

ENGINE — THE CALCULATION YOU MUST BUILD AND DEFEND

Every generated claim must resolve to a retrieved source span, with unsupported claims blocked and counted

VALIDATION TEST — HOW YOU PROVE IT IS RIGHT

The hallucination rate is measured with and without the verification layer, on the same question set

VIVA DEMONSTRATION, THREE MINUTES

Ask the same question with the layer off and on, and show the blocked claim

7. Submission Checklist

Work through this before you bind the report. Every line maps to a mark you can lose.

- The title on the title page matches the title printed against your names in section 6.
- The report runs to 25 to 30 pages in the prescribed structure.
- Times New Roman 12, headings 14, spacing 1.5, justified, one inch margins, spiral bound.
- Similarity index is 20 per cent or lower and you have the report to show.
- Every table and figure carries a number and a title.
- sources.csv is in the annexure and every source in it appears in the bibliography.
- Chapter 11 contains your own analysis, your validation result, and your interpretation of both.
- The assumptions file is written and both of you can defend every assumption in it.
- The repository is public, the README explains how to run the system, and commits span the project period.
- The application runs on a laptop you will carry to the viva, without needing the internet if possible.
- Your three-minute demonstration is rehearsed and includes one failure case.
- The presentation runs to eight to ten slides covering objectives, data sources, analysis and findings.
- Both hard copy and PDF are ready before the End Semester Examination.

> **One last thing.** The external examiner will pick one number from your report and ask you where it came from and how it was calculated. Prepare for that question on every number, not on the ones you expect to be asked about.